

Section A

Answer **all** the questions in this section.

- 1 Fig. 1.1 below shows 2 organelles from an animal cell.

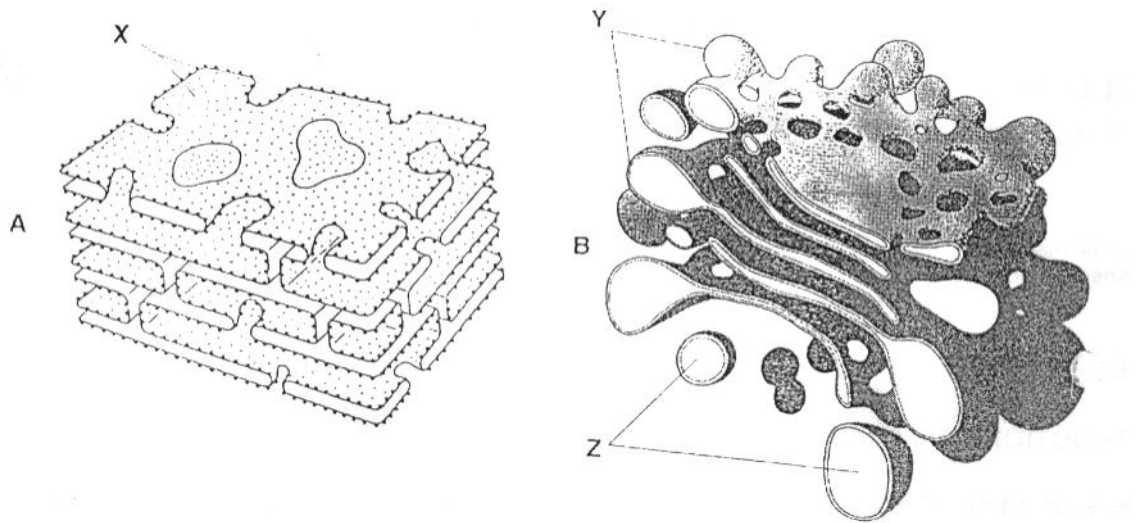


Fig. 1.1

- (a) (i) Name the Organelles A and B

A Rough Endoplasmic Reticulum

B Golgi apparatus

[1]

- (ii) Underline the cells in the following list that do not have any A.

liver cell; bacterium; virus; muscle fibre

[1]

- (iii) Name the structures Y and Z shown in Fig. 1.1B.

Y Golgi body / cisternae

Z Golgi vesicles/ Secretory vesicles

[1]

- (b) Explain the function of A and B with relation to protein synthesis.

Ribosomes on RER – site of protein synthesis ;

RER transports protein made by ribosomes through the cisternae ;

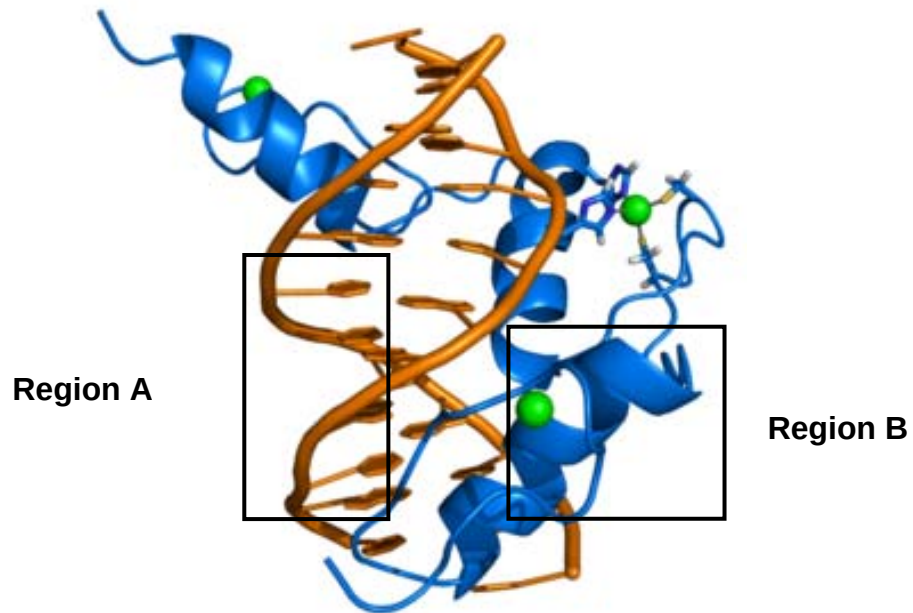
Golgi: Chemical modification e.g adding carbohydrate chain ;

Transport through golgi vesicles/ secretory vesicles to within cell/ out of cell

[4]

[Total: 7]

- 2 Fig. 2.1 illustrates the binding of a protein with a DNA molecule.



(a) With reference to Fig. 2.1,

(i) Describe in what ways the helical structure show in Region A and B differ.

Double strand (A) vs one single polypeptide (B);

For A each turn is 10 base pair, B is 3.6 amino acids per turn;

Bonds in A phosphodiester and hydrogen, B has peptide, ionic, hydrogen bonds;

[3]

Region B contains a zinc finger protein motif. It is a highly conserved protein motif and the binding specificity is conferred by a short stretch within the α -helix domain. Transcription factors often contain zinc finger motifs.

(ii) Suggest why a transcription factor only binds to certain part of a DNA molecule.

Transcription factor has binding site;

Complementary;

Different sequences of bases have different conformation/shape;

[3]

(b) (i) State what is a control element.

DNA in nature + regulate gene expression;

[1]

(ii) Explain how a control element can increase gene expression.

activator protein bind to enhancer;

Protein-protein interaction results in bending/ looping of DNA;

Association of enhancer-activator stabilises/ increases recruitment of transcription factors, leading to increased rate of transcription;

[2]

[Total: 9]

- 3 Fig. 3.1 below shows the movement of O_2 and CO_2 between two organelles, A and B, in a mesophyll cell of a leaf, in the presence of light.

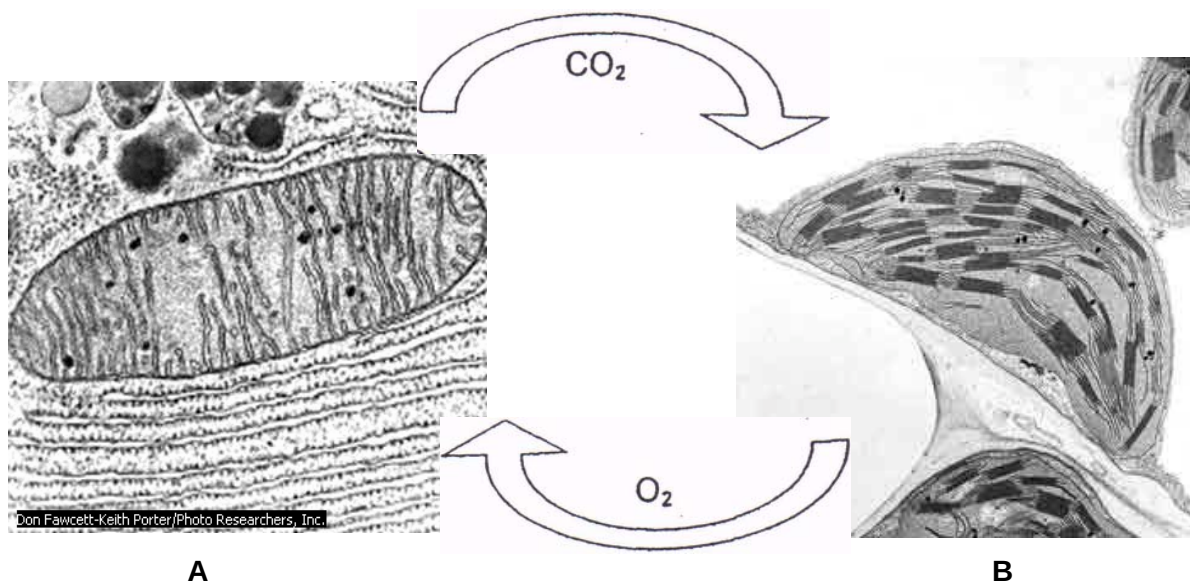


Fig. 3.1 Movement of O_2 and CO_2 between organelles A and B

- (a) State the process taking place in:

organelle **A**: Respiration/ Krebs cycle/ Oxidative Phosphorylation

organelle **B**: Photosynthesis/ Calvin cycle/ Photophosphorylation

[1]

- (b) Briefly describe how organelle B is structurally adapted to carry out its function/s.

Surface area of Thylakoid/grana increase for more chlorophyll pigments/protein complex for absorbing light (any description of light harvesting);

Membrane impermeable to H^+ (description of chemiosmosis);

Stroma contains enzyme/ DNA and ribosomes for series of enzyme catalysed reactions in Calvin Cycle;

Different photosystem pigments (PSI/ PS II), trap different wavelength of light;

[3]

- (c) ATP synthesis takes place in both organelles A and B. State three ways in which the process of ATP synthesis differs in the two organelles.

Mitochondria – electrons/protons from oxidation of organic compounds vs chloroplast from photolysis of water/chlorophyll a;

NAD/FAD vs NADP as electron carrier;

H⁺ released combine with oxygen to form water vs H⁺ recycled;

Oxygen required as final electron/hydrogen acceptor vs oxygen released by photolysis;

Electron from ETC (mito) combine with oxygen/ hydrogen ions to form water, electrons from ETC returned to chlorophyll molecules/ NADP;

[3]

[Total: 7]

- 4 A variety of oilseed rape shows variation in two characters: stem height and chlorophyll distribution in its leaves. Plants may be tall or short. Chlorophyll distribution may be uniform, producing normal green leaves; patchy, producing variegated leaves; or totally absent. Plants in which chlorophyll is totally absent die before they emerge above ground as seedlings.

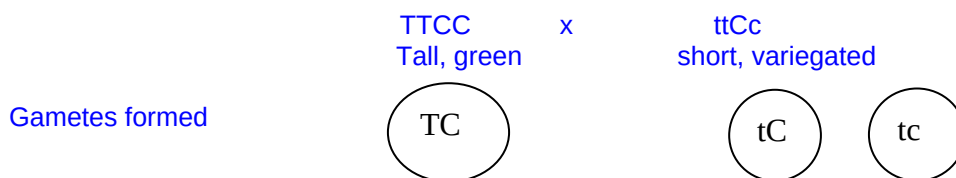
Pure breeding plants with tall stems and normal green leaves were crossed with plants with short stems and variegated leaves. The generation from this cross consisted of plants with tall stems and green leaves and plants with tall stems and variegated leaves in a ratio of 1:1.

- (a) Using suitable symbols, construct a diagram of this cross to show the genotypes of the parents and the offspring produced.

Let T represent allele for tall plants, t represent allele for short plants

Let C represent allele for normal chlorophyll distribution, c represent allele for absence of chl

Parental generation



By independent assortment

	tC	tc
TC	TtCC	TtCc

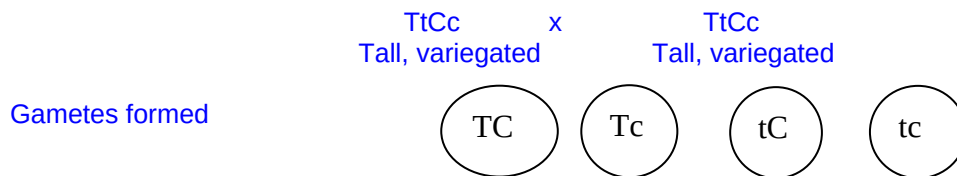
[3]

- (b) The first generation plants with tall stems and variegated leaves were self-pollinated and 1000 seeds sown. Six weeks after germination, the phenotypes of the second generation plants were recorded. The following results were obtained:

Plants with tall stems and green leaves	190
Plants with tall stems and variegated leaves	386
Plants with short stems and green leaves	61
Plants with short stems and variegated leaves	126

Construct a diagram to show the results of the self pollination.

Parental generation



By independent assortment

	TTCC	TTCc	TtCC	TtCc
	TTCc	<u>TTcc</u>	TtCc	<u>Ttcc</u>
	TtCC	TtCc	ttCC	ttCc
	TtCc	<u>Ttcc</u>	ttCc	<u>ttcc</u>

Tall, green	Tall, variegated	Short green	Short variegated
1 TTCC, 2 TtCC	2 TTCc, 4 TtCc	1 ttCC	2 ttCc
3 Tall, green	6 Tall, variegated	1 Short green	2 Short variegated
190 Tall, green	386 Tall, variegated	61 Short green	126 Short variegated

- 5 Restriction enzymes are used during genetic engineering. The target sites for *EcoRI*, *BamHI* and *AluI* is shown in Fig. 5.1 below.

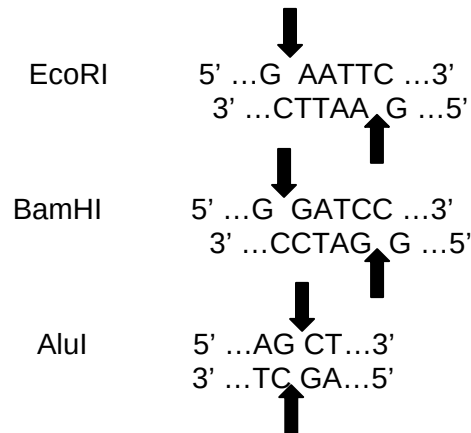


Fig. 5.1

A bacterial plasmid is shown in Fig.5.2 below. There are no target sequences for the three restriction enzymes in the rest of the plasmid.

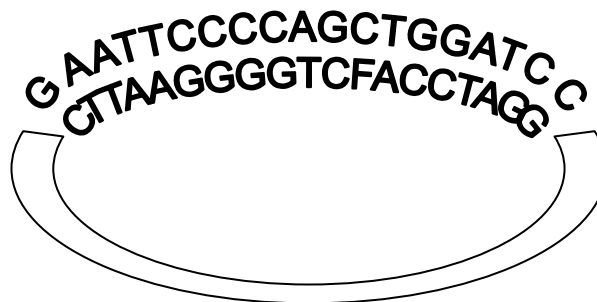


Fig. 5.2

With reference to **Fig. 5.1** given above,

- (a) State how many fragments of DNA will be present after the plasmid has been cut with

- (i) a mixture of *EcoRI* and *BamHI*

2 fragments

- (ii) a mixture of *EcoRI* and *AluI*

2 fragments

[2]

- (b) A student incubates the plasmid molecule shown in Fig. 5.2 with *EcoRI* to insert a human gene. Outline how restriction enzymes may be used in the formation of recombinant DNA molecules.

Use same restriction enzyme to cut both plasmid and foreign gene;

Incubate plasmid and foreign gene together, anneal by CBP/H bonding;

Ligase added, catalyse formation of phosphodiester bonds;

[3]

- (c) Name two other enzymes used in the formation of recombinant DNA and briefly describe their roles.

terminal transferase: catalyse addition of nucleotides to blunt ends;

reverse transcriptase: catalyse formation of DNA molecule from a RNA template;

ligase: catalyse formation of phosphodiester bonds;

[2]

- (d) Gel electrophoresis is usually performed after restriction digestion to separate DNA fragment. Briefly describe how DNA fragments may be separated by gel electrophoresis.

fragments separated based on size and charge;

DNA loaded in well at negative end of electrode, DNA is negatively charged and migrates to the positive electrode/ anode;

Rate of migration is inversely proportional to size of fragment / appropriate description of speed of fragments + relative position of fragment reference to well position;

[3]

[Total: 10]

Section B

Answer **EITHER** 6 or 7

- 6 (a) Describe the structure and function of portions of eukaryotic DNA that do not encode for protein or RNA. [10]

1. introns found within genes;
2. transcribed into RNA but not translated into protein;
3. spliced out before translation;
4. promoted the evolution of new proteins by duplication of exons or repositioning of exons within a gene;
5. contributed to alternative splicing
6. control elements – non-coding DNA which regulate gene expression;
7. promoters – near a structural gene,
8. allows RNA polymerase to bind & initiate basal transcription;
9. enhancers : distal, upstream of gene;
10. allows activators to bind to increase rate of transcription;
11. silencers : allows repressors to bind to decrease rate of transcription;
12. centromeres made up of tandem repeats;
13. centromere function to hold sister chromatids together;
14. associated with kinetochore to
15. allow binding to chromatids;
16. allows separation of sister chromatids during nuclear division;
17. telomeres : ends of chromosome;
18. made of tandem repeats;
19. function to protect ends of chromosome;
20. prevent abnormal fusing of chromosomes;
21. intergenic sequences : DNA sequences between structural genes;
22. contains repeated sequences;
23. scattered about genome;

(1/2 mark per point)

- (b) Describe how a gene may be transferred into a named bacterium to obtain large

amounts of useful proteins.

[10]

1. Isolation of human gene; R: isolate DNA
 2. Named: E. coli; underline, large caps/small caps
 3. eukaryotic gene such as insulin contains introns and exons.
 4. E. coli unable to carry out post-transcriptional modification;
 5. so cDNA must be used so it contains only exons/ amino acid sequence used to synthesize artificial insulin gene without introns;
 6. many human proteins contain more than one polypeptide. E.g. insulin contains α -chain and β -chain. But E. coli cannot carry out post-translational modification;
 7. PCR carried out to increase number of copies;
 8. genes for α -chain and β -chain are cloned into different plasmids
 9. vector plasmid must contain 2 selection marker genes of which one can be cut by a restriction enzyme;
 10. e.g. ampicillin resistance marker and tetracycline resistance with multiple cloning site which restriction enzyme cuts;
 11. contains a highly active bacterial promoter at 5' of human gene;
 12. cut with human gene and plasmid vector with same restriction enzyme; R: treat
 13. results in complementary sticky ends;
 14. Incubate together and treat with ligase which forms phosphodiester bonds;
 15. (Recombinant plasmid)s with human gene inserted and (non-recombinant plasmid) with reannealed sticky ends are formed;
 16. Transformation or description : uptake of naked DNA from the surrounding by bacteria; R if contradicting information present
 17. Induce transformation by treating with CaCl_2 and heat shock/ by electroporation
 18. Temporary gaps in membrane of E. coli appear allows plasmids to enter;
 19. Selection : identification of transformed bacteria containing recombinant DNA;
 20. Grow E. coli on medium with ampicillin + only successfully transformed E. coli will survive;
 21. Insertional inactivation of tetracycline occurs in recombinant plasmids which human gene interrupts tetracycline resistance gene/ recombinant plasmids containing human gene is not resistant to tetracycline;
 22. Replica plating;
 23. Colonies which survive ampicillin but not tetracycline are picked and cultured;
 24. Different colonies of E. coli will produce α -chain or β -chain of insulin depending on which gene is inserted.
 25. bacteria lysed and α -chain or β -chain subunits/polypeptides are harvested separately;
 26. polypeptides treated to cleave off N-formyl-methionine, a feature of prokaryotic translation;
 27. cleaved α -chain and β -chain subunits incubated together under conditions to induce formation of disulphide bond to form quaternary structure;
- (1/2 mark per point)**

[Total : 20]

- 7 (a) Discuss the benefits and ethical concerns of the Human Genome Project. [10]
1. provides a detailed physical, genetic map of the human genome and DNA sequence of entire human genome
 2. allow elucidation of genes associated with human diseases/ understanding genetics basis of disease;
 3. allows early detection or diagnosis via genetic testing;
 4. e.g risk of breast cancer, diabetes and colon cancer;
 5. improve diagnosis diseases prenatally;
 6. pharmacogenomics : design customized drugs based on genetic profile;

7. enables novel treatment through gene therapy as gene sequences are now readily available;
8. pharmaceutical products such as medicine can be synthesized using readily available gene sequences; e.g. humulin and tissue plasminogen activator (TPA) given to after heart attack to dissolve blood clots;
9. a means of risk assessment to evaluate health risks of individuals who may be exposed to radiation or carcinogens;

max : 6 points

10. ethical issues of pre-implantation genetic testing viewed as “tampering with nature” by certain religious groups;
11. To-be parents may have to make difficult decisions of terminating pregnancy due to presence of genetic disorder. It may amount to murder since there are different definitions of when a foetus is recognized as living;
12. positive test result indicates that a person possesses a known mutation and may not necessarily develop cancer. Are the general population informed adequately;
13. management of genetic information, who owns the right to the genetic information? The person sample taken from or the hospital or company who sequenced the information?
14. work-related discrimination: is it right for employers to demand DNA testing because it can reveal any possible high risk disease the employee may develop and thus increase costs of employing this person;
15. insurance-related discrimination: is it right for insurance companies to demand DNA testing because it can reveal any possible high risk disease which is not covered under the insurance policy;
16. Designer drugs developed by pharmacogenomics discriminates against the poor as only the wealthy have the ability to pay for such services;

Max: 4 points

(b) Describe the different types of stem cells and their functions.

[10]

1. properties of stem cells : unspecialized / undifferentiated;
2. can give rise to other cells via mitosis;
3. self-renewal;
4. 2 daughter cells one remains unspecialized, the other one differentiate;
5. types of stem cells : embryonic SC obtained from blastocyst / eight-cell stage;
6. Function : cells develop into tissues which give rise to specialized cell types that make up the whole organisms;
7. umbilical cord stem cells : found in blood of umbilical cord of new born;
8. adult stem cells :stem cells found in adult tissues such as bone marrow, brain, blood vessels, skin and liver;
9. Function: to generate replacement cells lost through disease, tissue injury and wear and tear;
10. e.g. in bone marrow, haematopoietic cells form all types of blood cells;
11. e.g. spermatogonial stem cells
12. e.g. neural stem cells
13. potency : AVP;

[Total : 20]

END OF PAPER